

- If $f(x,y) = f_x(x) f_y(y)$ for all (x,y) , then X and Y are independent RVs.
- If $p(x_i, y_j) = p_x(x_i) p_y(y_j)$ for all (x_i, y_j)

$$\rho = \frac{E(XY) - E(X)E(Y)}{\sqrt{E(X^2) - (E(X))^2} \sqrt{E(Y^2) - (E(Y))^2}}$$

$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$

$$\rho = \frac{\frac{1}{n} \sum_{i=1}^n x_i y_i - \left(\frac{1}{n} \sum_{i=1}^n x_i\right) \left(\frac{1}{n} \sum_{i=1}^n y_i\right)}{\sqrt{\frac{1}{n} (\sum_{i=1}^n x_i^2 - n \bar{x}^2)} \sqrt{\frac{1}{n} (\sum_{i=1}^n y_i^2 - n \bar{y}^2)}}$$

$$r = \frac{\sum_{i=1}^n x_i y_i - \frac{\sum_{i=1}^n x_i \sum_{i=1}^n y_i}{n}}{\sqrt{\sum_{i=1}^n x_i^2 - \frac{(\sum_{i=1}^n x_i)^2}{n}} \sqrt{\sum_{i=1}^n y_i^2 - \frac{(\sum_{i=1}^n y_i)^2}{n}}}$$

